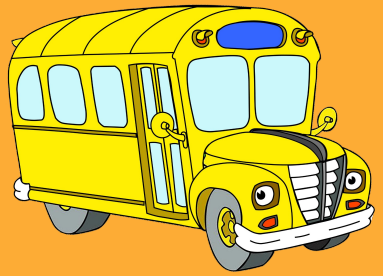
Cellular Respiration

brought to you by:
Magic School Bus Tour Company



Hey guys! My name is Michelle, and I'm going to be your tour guide and bus driver for the Cellular Respiration Tour at the Magic School Bus Tour Company!

Before we get started, everyone take a quick look at your travel guide brochures right now so you can get a better idea of what we'll be expecting in a bit! Please familiarize yourself with what we are about to do, both for educational purposes and liability issues. Please don't ask me why, I'm just the bus driver, and the one responsible for your lives. But no, I don't know why, it's just company policy.

CELLULAR RESPIRATION TOUR

Cellular respiration is the **catabolic pathway** that breaks down **glucose** to **create ATP** in the presence of **oxygen**, and that's what we will be observing today inside a cell in a real human!

Metabolism is the term used to describe all the chemical reactions involved in maintaining the life of an organism. It's a cycle of **catabolic and anabolic pathways or reactions**. Different types of energy-containing nutrients like carbohydrates, fats, and proteins go through catabolism and create **energy-depleted products** like CO2, H2O, and NH2. Precursor molecules like amino acids, sugars, fatty acids, and nitrogenous bases go through anabolism to make different types of cell **macromolecules** like proteins, polysaccharides, lipids, and nucleic acids.

Catabolic pathways break down large, complex molecules into smaller, simpler products to **release energy**, so the products have **less potential energy** than the reactants because energy is **released** when bonds are broken. Anabolic pathways **consume energy** in order to **build** large, complex, molecules from smaller monomers, so the products have more potential energy than the reactants because energy is **invested** into the reaction.

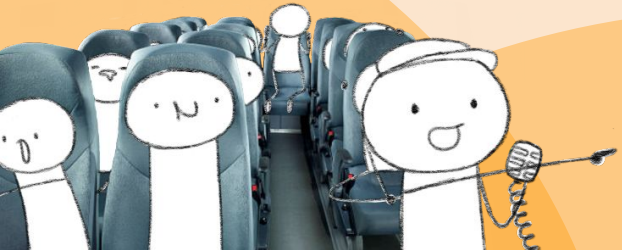
We are going to go through glycolysis, the Krebs cycle, and then see the electron transport chain from afar!

CHECKPOINTS

1. ***CYTOSOL***
2. ***MITOCHONDRION***

To get in the cell, we are going to lodge ourselves into this piece of gum as it goes down our unsuspecting and definitely consenting friend's throat, because contrary to popular belief, gum does get digested and will not remain in your gut for seven years. Seatbelts everyone!

Here we gooooo!!!!!!!!!!!



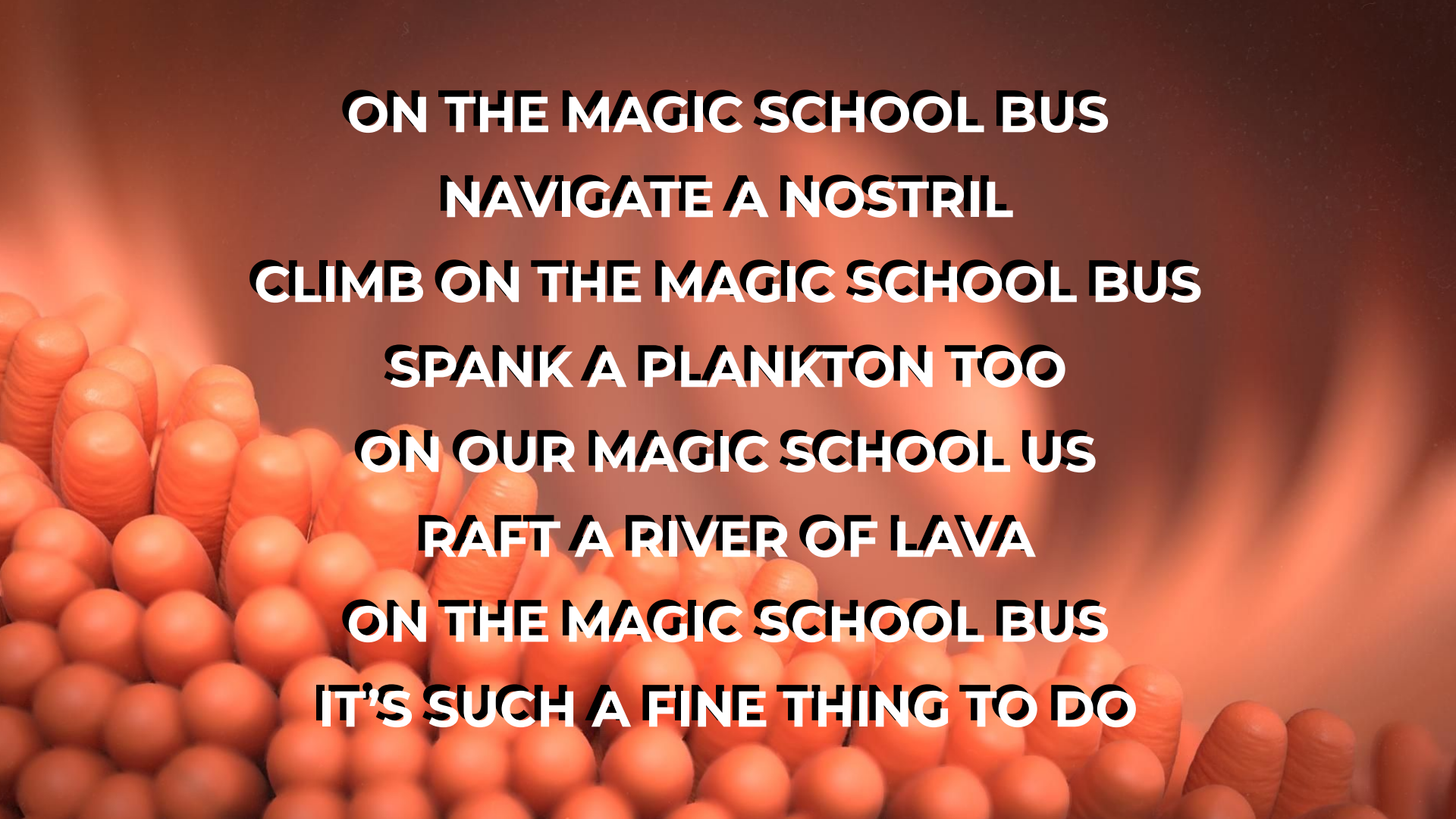
CRUISIN' DOWN ON MAIN STREET, YOU'RE RELAXED AND FEELING GOOD
NEXT THING THAT YOU KNOW, YOU'RE SEEING...
AN OCTOPUS IN OUR NEIGHBOURHOOD?

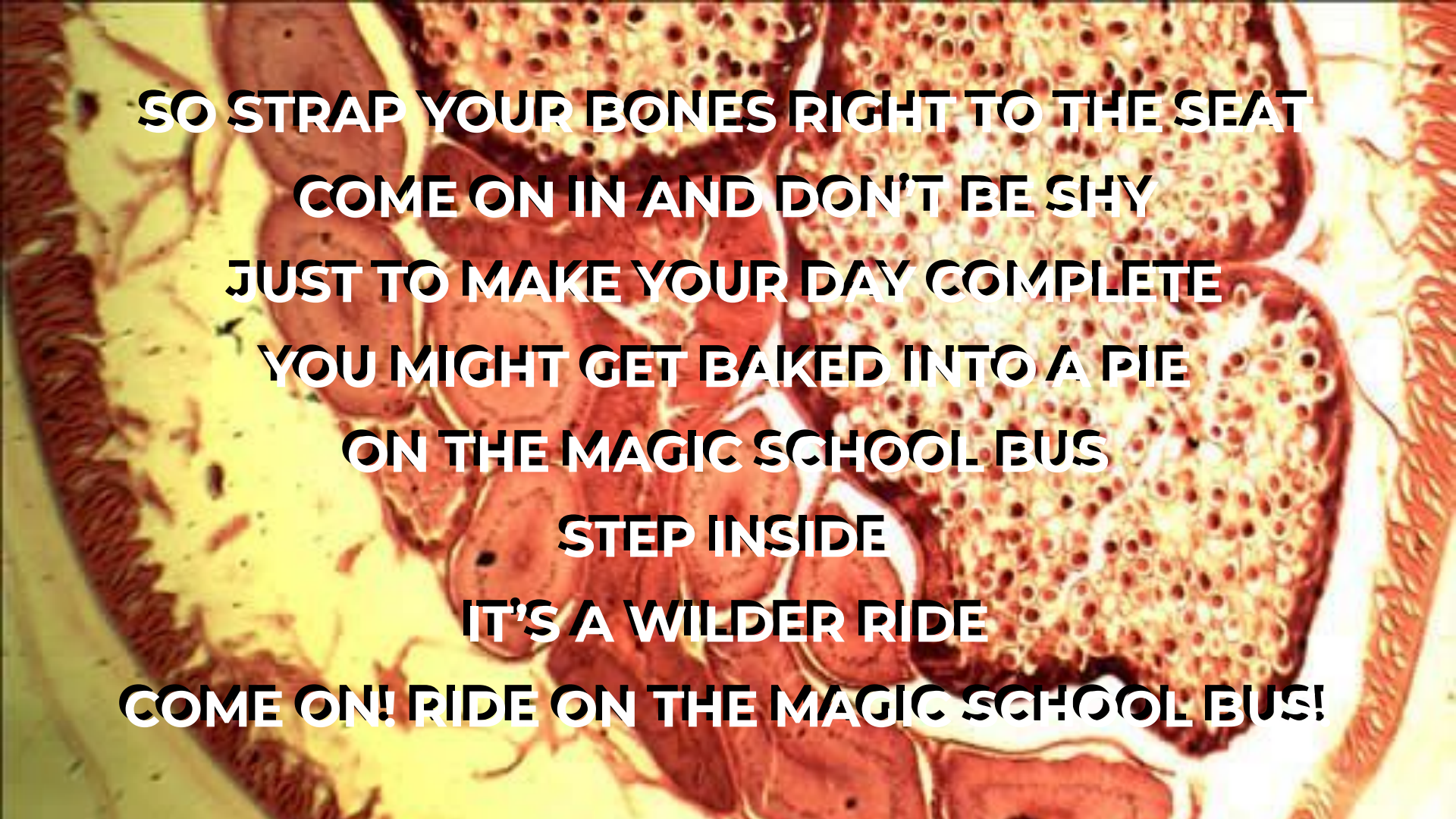
SURFING ON A SOUND WAVE

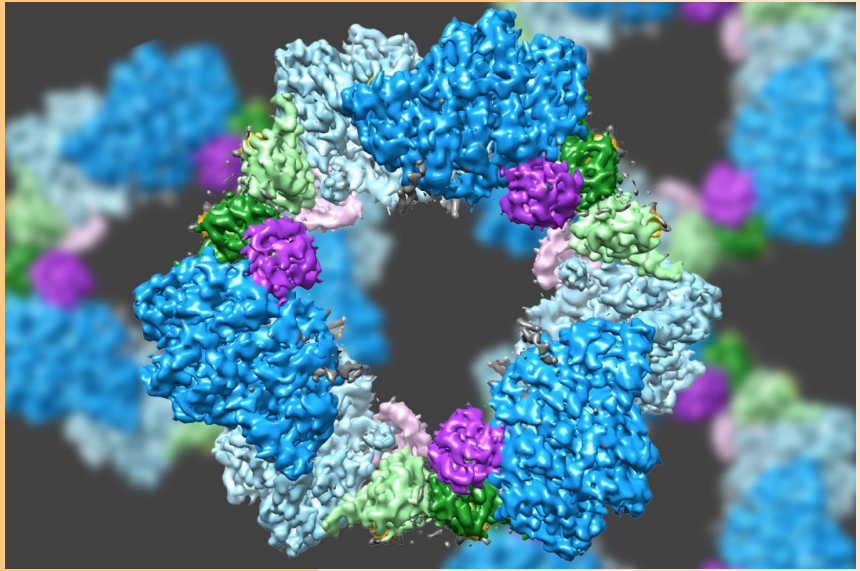
SWINGING THROUGH THE STARS

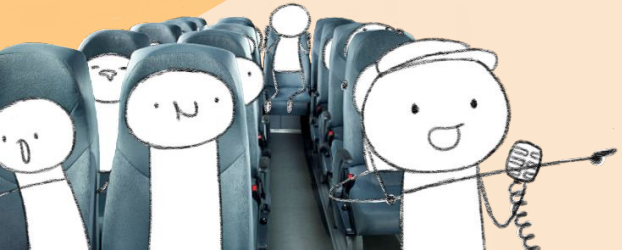
TAKE A LEFT AT YOUR INTESTINE

TAKE YOUR SECOND RIGHT PAST MARS


ON THE MAGIC SCHOOL BUS
NAVIGATE A NOSTRIL
CLIMB ON THE MAGIC SCHOOL BUS
SPANK A PLANKTON TOO
ON OUR MAGIC SCHOOL US
RAFT A RIVER OF LAVA
ON THE MAGIC SCHOOL BUS
IT'S SUCH A FINE THING TO DO

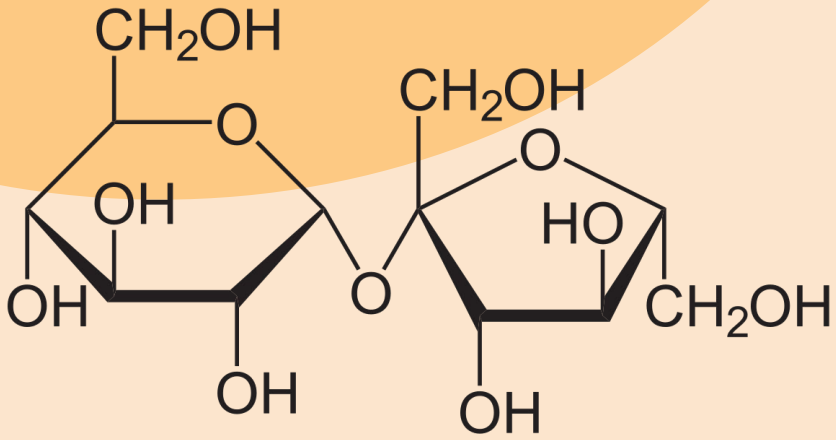

SO STRAP YOUR BONES RIGHT TO THE SEAT
COME ON IN AND DON'T BE SHY
JUST TO MAKE YOUR DAY COMPLETE
YOU MIGHT GET BAKED INTO A PIE
ON THE MAGIC SCHOOL BUS
STEP INSIDE
IT'S A WILDER RIDE
COME ON! RIDE ON THE MAGIC SCHOOL BUS!





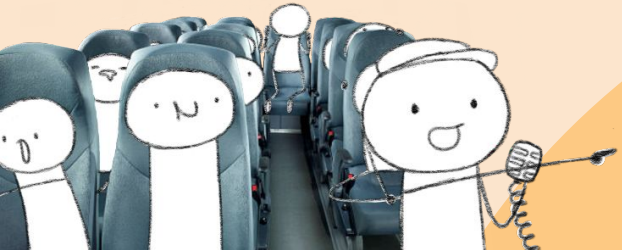
As we approach the **pyloric sphincter**, we are going to shrink ourselves even further and attach to these little **sugar molecules** that have been **broken down** from the gum by the **enzymes**, and into the small intestine we go, to be **absorbed** by the body!

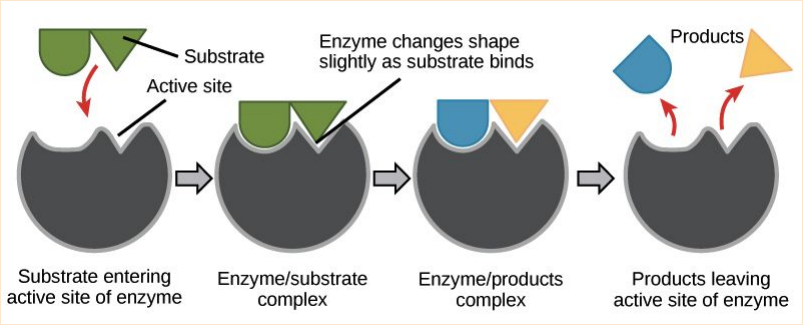
Let's take a look at what's happening right now to our sugar molecule! **Enzymes**, also known as **catalysts**, in short, are proteins that help **chemical reactions** run a lot more **smoothly, precisely, and quickly**, kinda like your HR managers or Quality Control guys in your offices! I wouldn't know, I don't work a boring desk job.

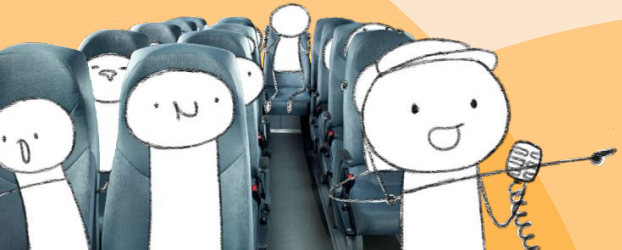


The fun thing about enzymes is that they can be **reused** after each reaction, because they **aren't actually used up** in the reaction. They just help it go **quicker**, the enzyme itself is not chemically involved and its **composition will not change**.

The sugar molecule we are currently riding on is called **sucrose**, and you might know it as your regular **table sugar**. It consists of a **glucose** molecule and a **fructose** molecule bonded together.

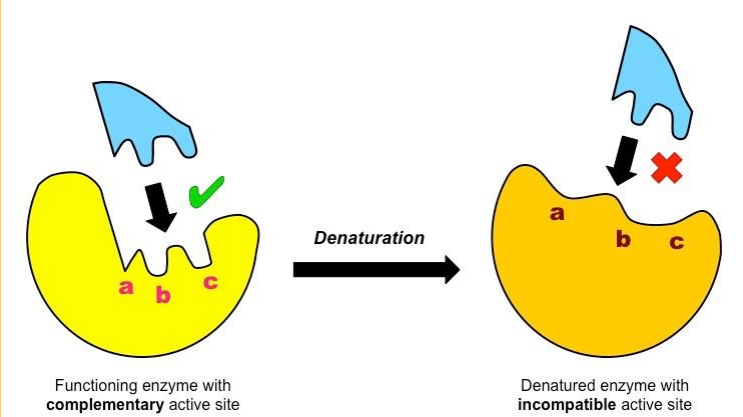


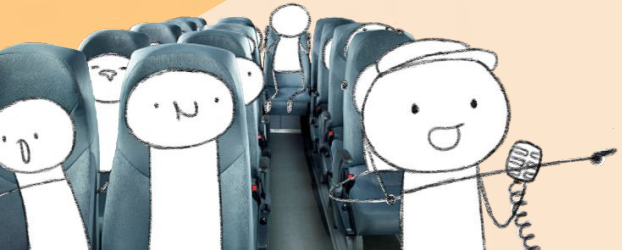




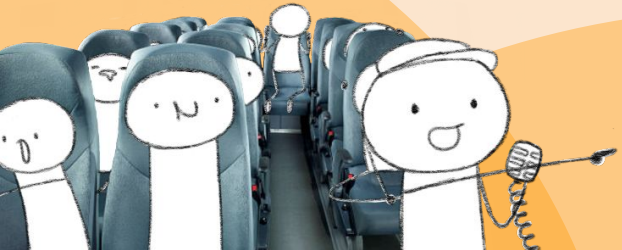
As we slowly approach the enzyme, you can see that the sucrose molecule, or as we should call it, the **substrate**, fits perfectly into the **active site**. The substrate **binds** to the enzyme, and we've now formed an **enzyme-substrate complex**.

This binding places **stress** on the glucose-fructose bond due to **charged active sites** that repel or attract certain parts of the substrate, distorting and straining the bond and **breaking** it!



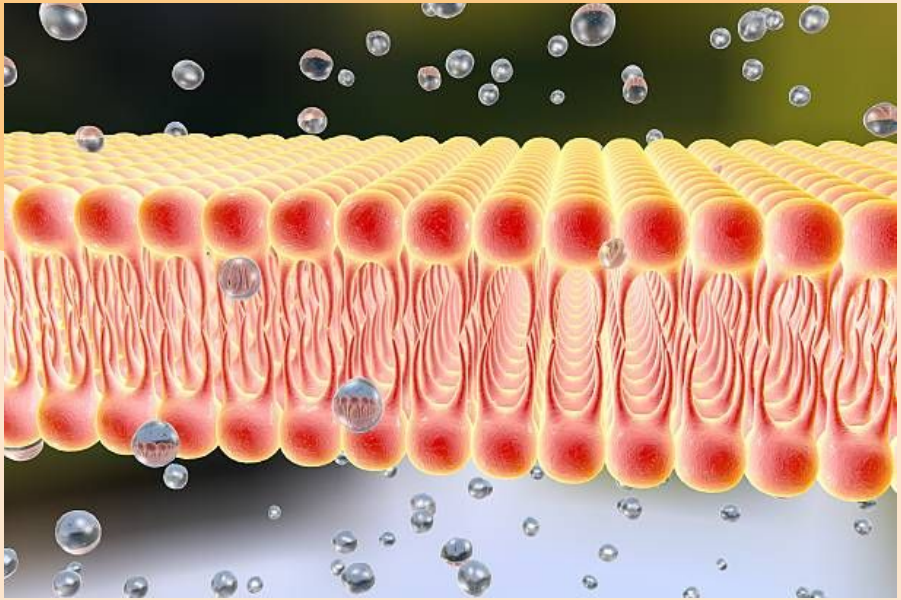


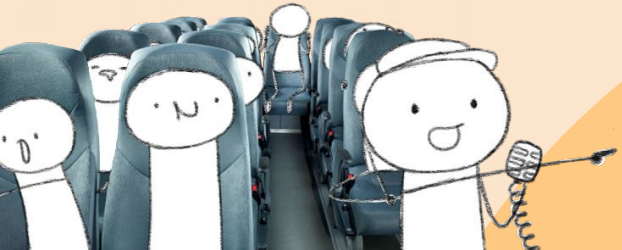
Each enzyme binds to a specific substrate and is responsible for catalyzing a **specific** reaction. They're our favourite jack of no trades, master of one! However, they're also little divas and can only function in **certain temperatures** and at **certain pHs**. However, if the conditions are not just right, the enzymes will **denature** and change shape, and it will no longer be able to carry out its job and bind to the substrate properly. Still, we love them because they allow our metabolic reactions to happen smoothly, quickly and efficiently, and we would not be able to-



Woo! The bond broke! I'll bet you didn't expect that. We've been freed from the fructose molecule, so it's going to go on its own path and we're going to follow this glucose molecule into Cell City! ¡Vámonos!

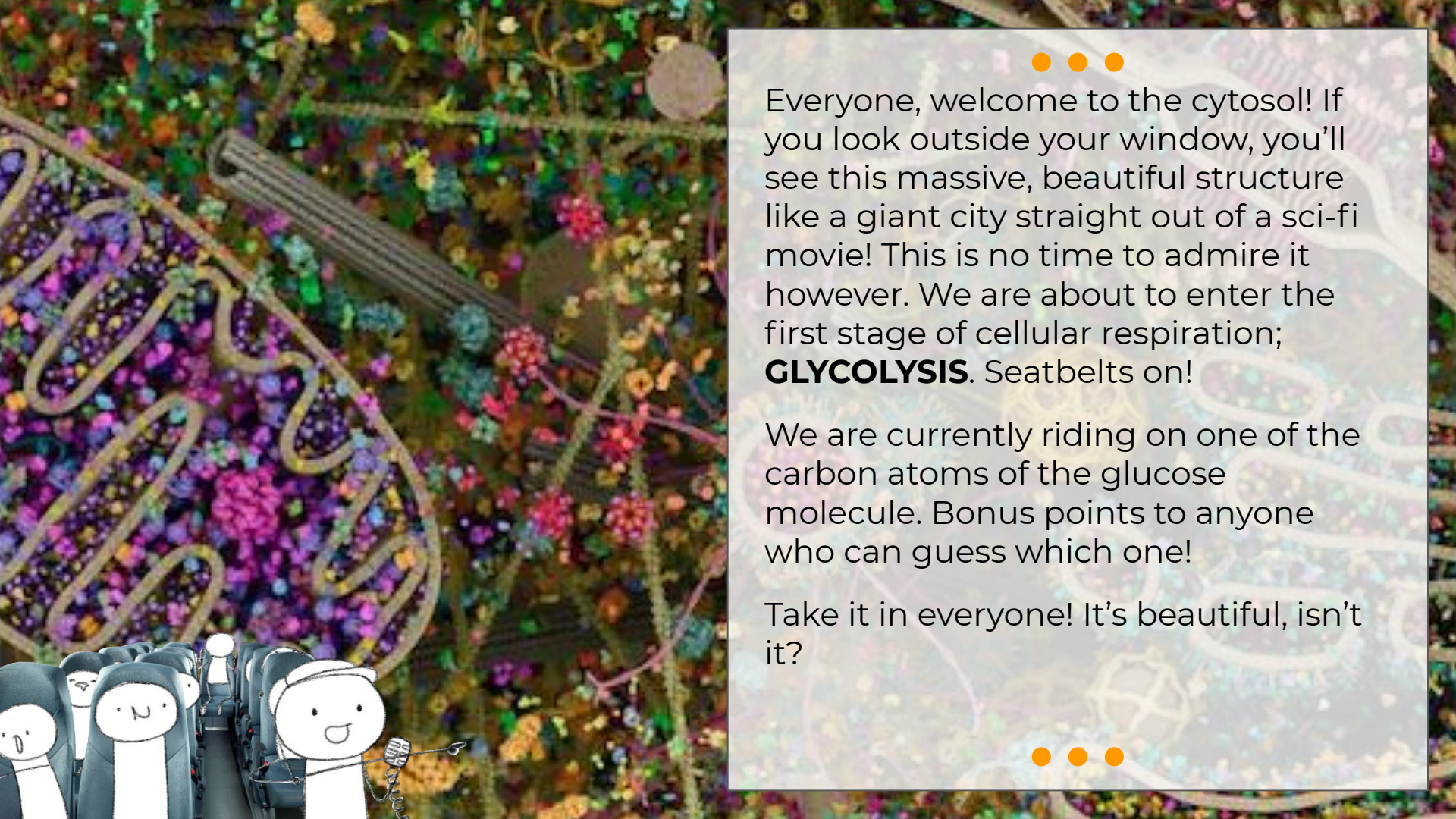
Nickelodeon please do not sue me.





Ding ding ding ding ding! Wake up! Wake up! If you don't wake up right now you're not gonna get your money's worth, and that would honestly be terrible. We are currently riding on a glucose molecule and we're about to touch down in this little field of **phosphate heads** you see right here called the **cell membrane**.

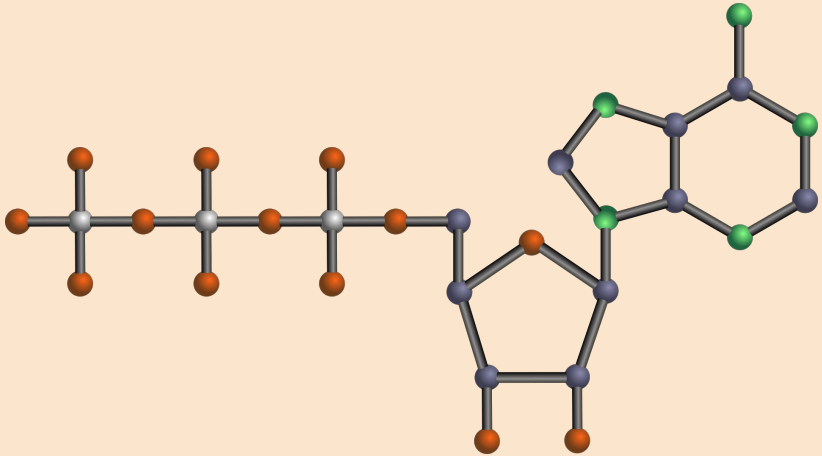
As we near the touchdown... oh, what's this? We are going right through the cell membrane through **facilitated diffusion**, because glucose is such a large molecule and can't just slip right in. It's like we're being swallowed right into another tinier body!



Everyone, welcome to the cytosol! If you look outside your window, you'll see this massive, beautiful structure like a giant city straight out of a sci-fi movie! This is no time to admire it however. We are about to enter the first stage of cellular respiration; **GLYCOLYSIS**. Seatbelts on!

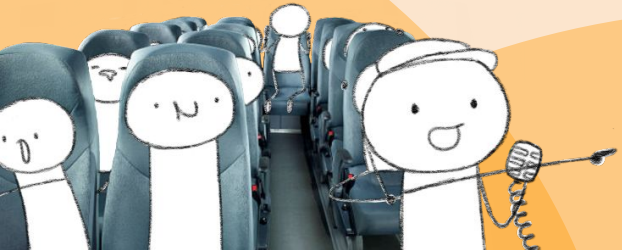
We are currently riding on one of the carbon atoms of the glucose molecule. Bonus points to anyone who can guess which one!

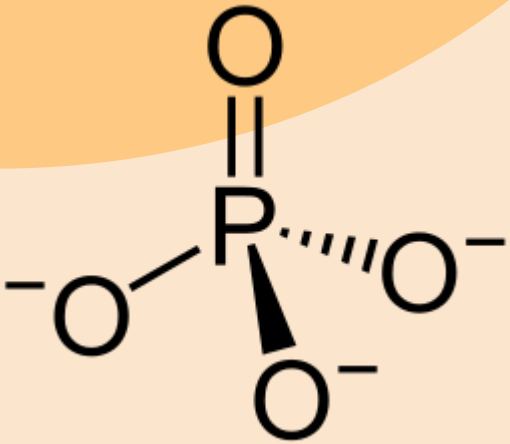
Take it in everyone! It's beautiful, isn't it?



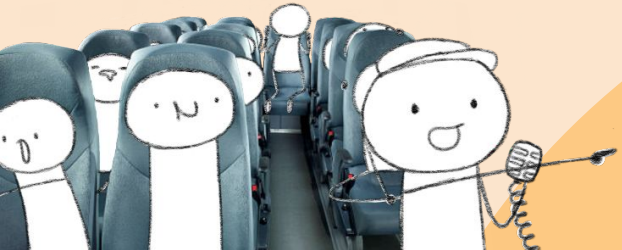
Thousands of ATP are produced in milliseconds. All around us are thousands of glucose molecules being mercilessly ripped apart by this natural phenomena we can only describe as life!

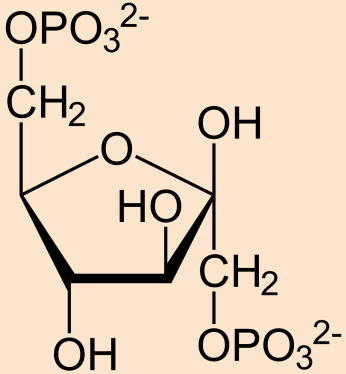
Look! Right there! That carbon pentagon connected to an adenine and bunch of phosphate groups! That's our friend **ATP - adenosine triphosphate**. What, you guys think this earthquake is just happening for free? No way!

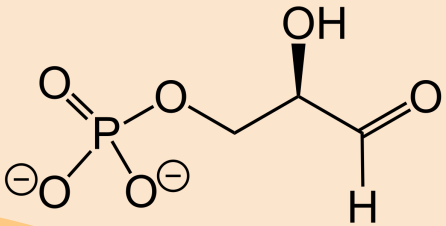


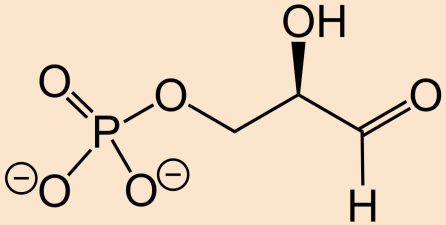


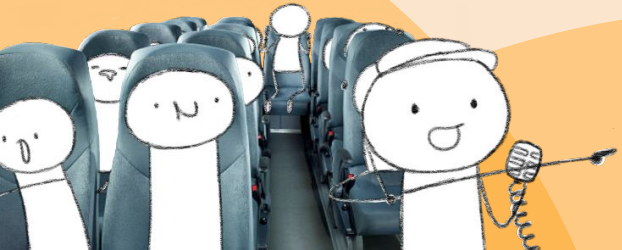
In glycolysis, ATP is invested into the process, which begins our **energy investment phase** of glycolysis, because the ATP is investing in us just like you should invest in my new company. You can see the **phosphate groups** breaking off and transferring to the monosaccharides. Look, there's two heading right for us! This addition of the phosphate group is called **phosphorylation**, and it's about to happen to us. Everyone count down with me! 3... 2... 1...





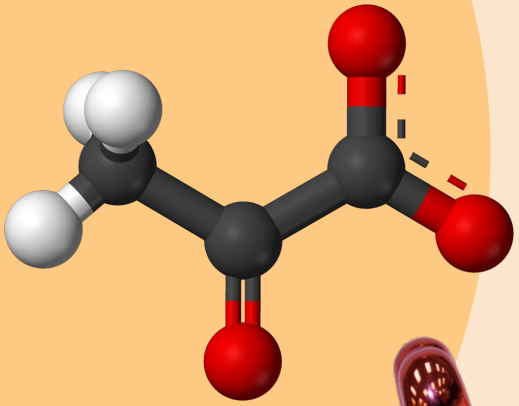


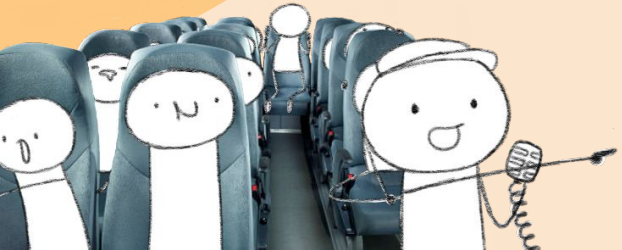




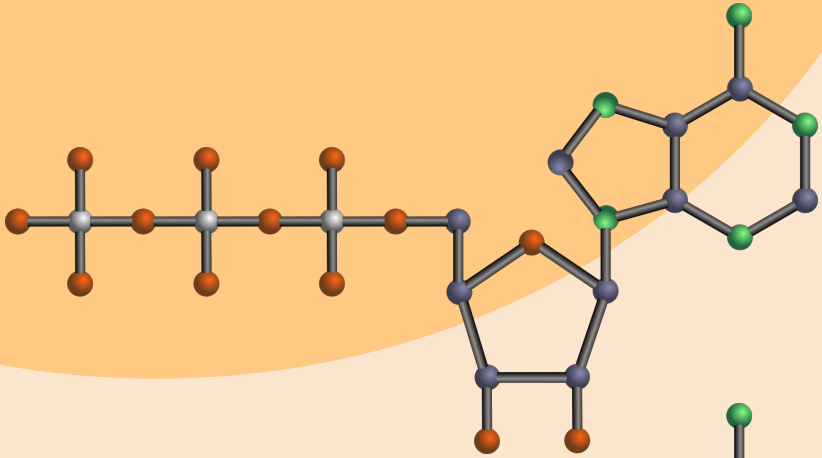
And now, as our ATP friends leaves, we can't call them ATP anymore. They are **ADP** now. Goodbye ADP! As for our glucose friend? This is definitely not glucose anymore. This is our friend **fructose-1,6-biphosphate**! If you haven't buckled up because you think you're cool, better do it now!

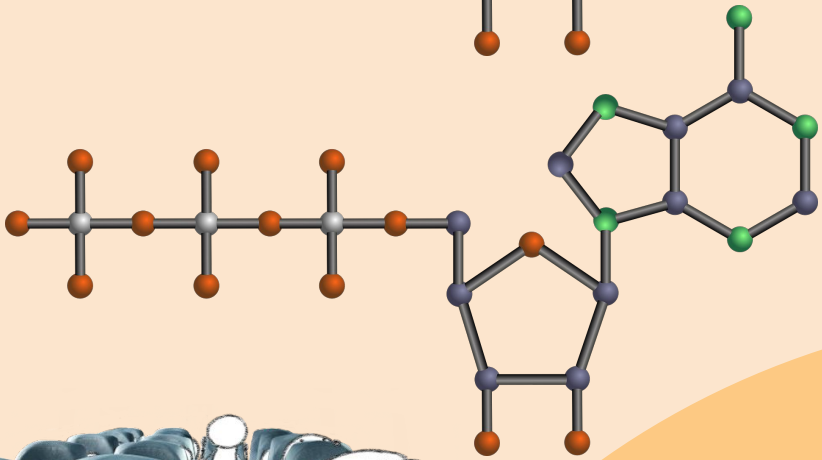
Fear not! There's no danger here.We can return to a normal size at any time we choose, we are completely safe, except for our friend who we are inside right now, they would not be too safe. Right now, we are being broken into 2 **glyceraldehyde 3-P** molecules. What a mouthful!

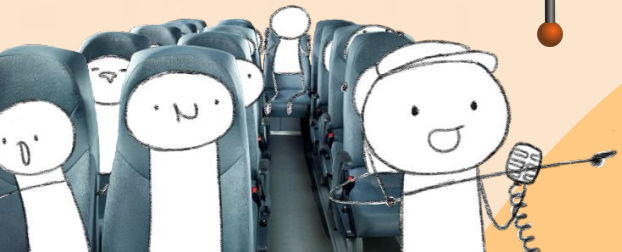




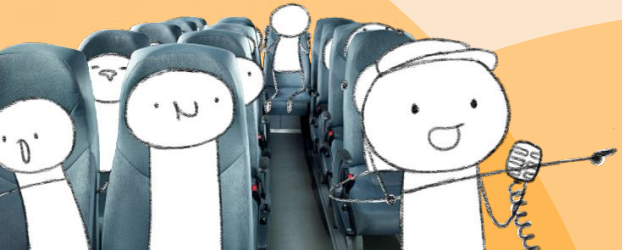
Now that the energy investment phase is over, we begin the **energy payoff phase**. Another phosphate group is moving towards at the moment, say hi phosphate group! And now, we are going to form **1,3-biphosphoglycerate**! Remember, there's **two** of us, so there's another 1,3-biphosphoglycerate somewhere out there who came in with us on the same glucose molecule, because each glucose molecule is eventually broken down into **2 pyruvate molecules**!



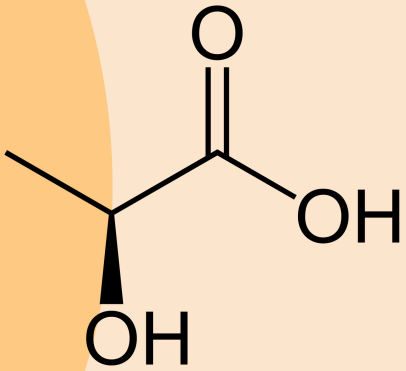


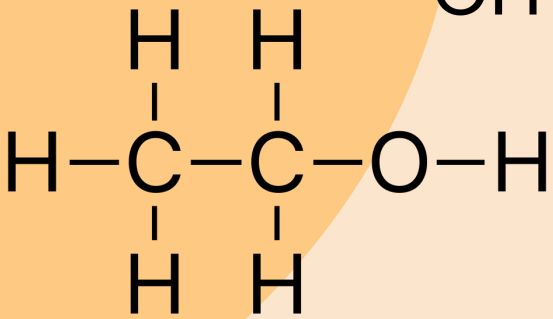


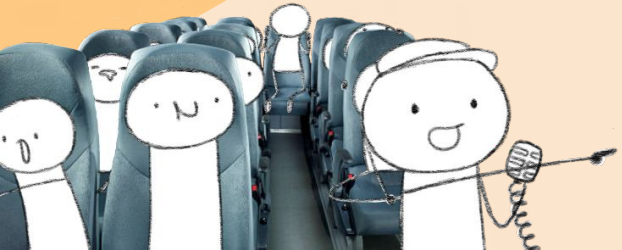
Now the **NAD+** is approaching us and steals our **H+ ions and electrons**? How rude! This is what they mean when they say that the tours are so expensive they're practically robbing you! And there it is, leaving as **NADH**. Here come 2 more ADP and they're going to steal our 2 phosphate groups away! And there they go, **2 ATP** have just been **produced**! Now since there's another half of the glucose molecule out there somewhere, this process has happened with that particle as well. The investment was only 2 ATP resulting in a **net production of 2 ATP** from one glucose molecule so far!



Since humans are oxygen-breathing mammals, cellular respiration in our bodies is an **aerobic process**, which is what happens in the **presence of oxygen**. Other organisms sometimes don't have oxygen, so they perform **anaerobic respiration**, which is a process that occurs in the **absence of oxygen**. In those cells, after glycolysis happens and they get their 2 pyruvate molecules, instead of going to the mitochondrion for the next step, we would have to stay in the **cytosol** to be fermented. **Fermentation** allows the **oxidation of NADH**, allowing glycolysis to continue.



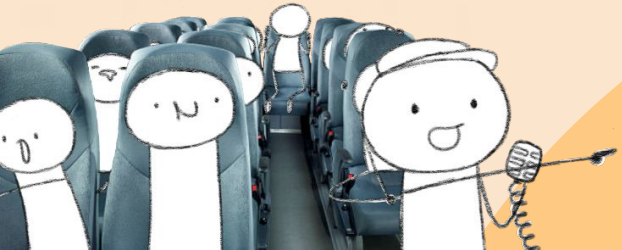


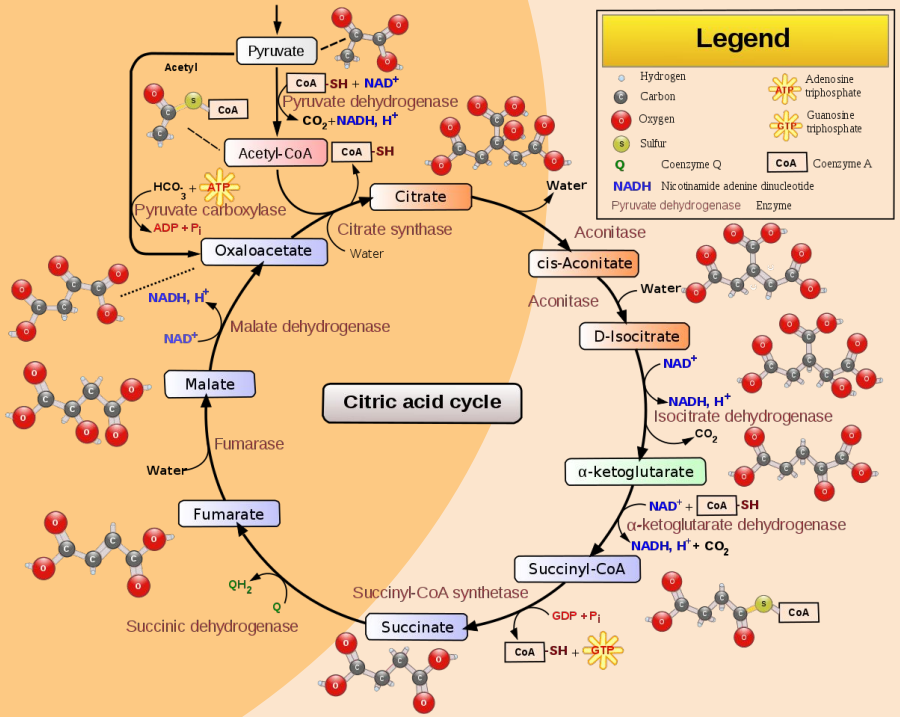


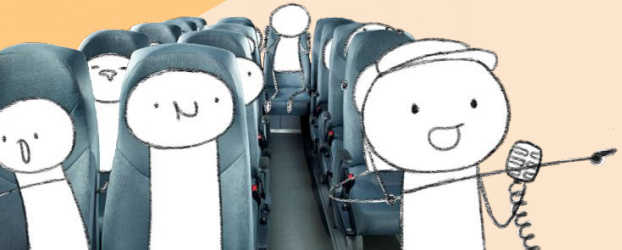
There are two types of fermentation: **alcoholic fermentation** and **lactic acid fermentation**, producing ethanol and lactic acid respectively. In alcoholic fermentation, pyruvate and NADH are invested, **acetaldehyde** oxidizes the NADH, and it produces CO2, **ethanol**, and NAD+. In lactic acid fermentation, **pyruvate** is the one oxidizing the NADH, and it produces **lactic acid** and NAD+.

Details aside, everyone look around. We have now completed glycolysis? How fun was that? No time for breaks though, because look outside. What do you see? Oxygen. A very friendly molecule, especially as a human. As a human studying glycolysis, in school, however? All the adults know what we're about to do. We are gonna say hi to a old enemy of ours. That's right, it's time to go through...

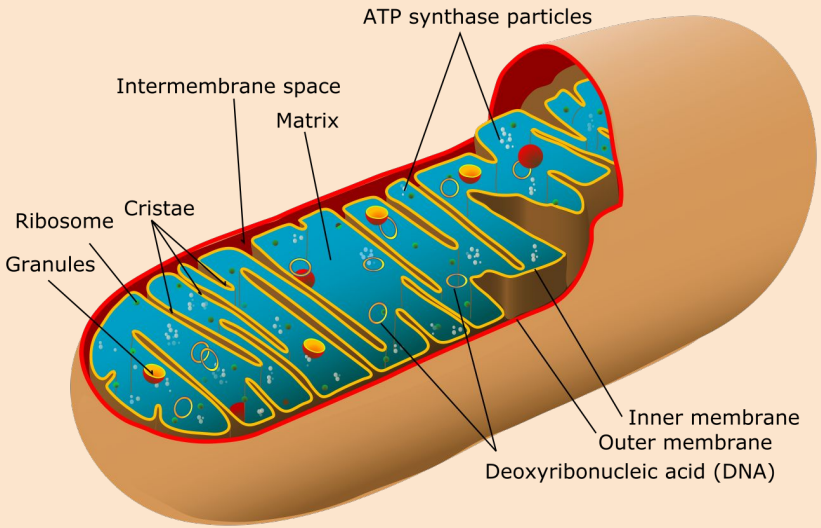
THE Krebs Cycle... ooooooooo

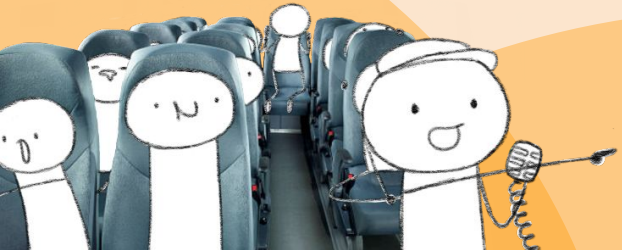






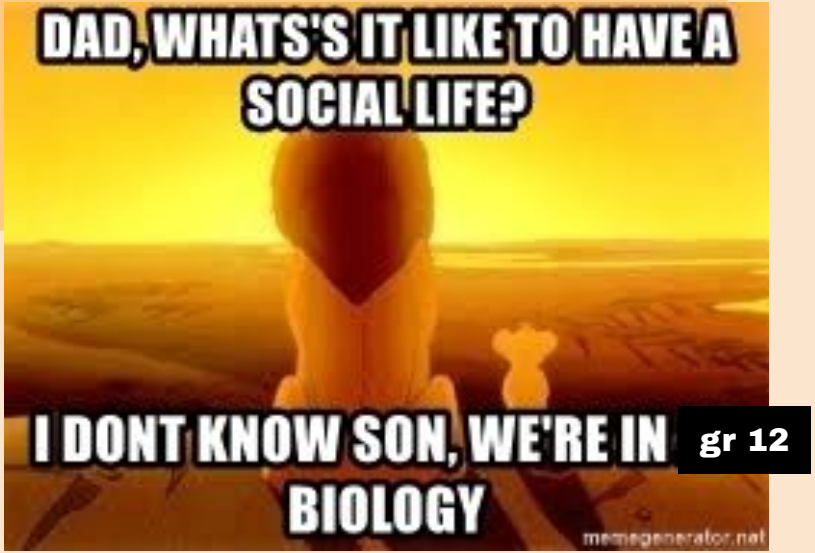
Giant bean approaching! This is what we call the **mitochondria**, a.k.a. the powerhouse of the cell! Everything else will be in the mitochondria, including the Krebs Cycle, also known as the **Citric Acid Cycle**, which you will have the privilege of experiencing in just a moment! This is an **aerobic process** and thus will not function without oxygen, even though the cycle itself **does not use oxygen**.





Everyone take a look at your map real quick and find the mitochondria. These little folds inside it are called **cristae**, and they increase the **surface area** so that more cellular respiration reactions can take place! There's also **outer membranes, intermembrane spaces, and inner membranes**, but we're headed for the **matrix**, where **pyruvate oxidation** will take place.

This pyruvate molecule still has a lot of energy, so what the Krebs Cycle is gonna do is finish breaking these molecules down. They will be completely oxidized and the carbon molecules will be released as **CO2**!



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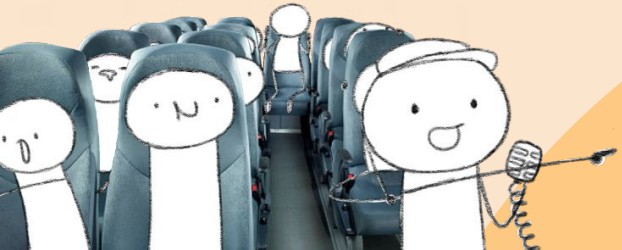
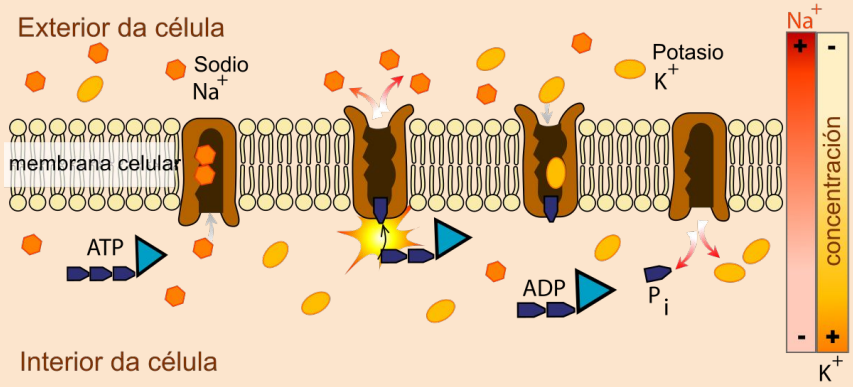
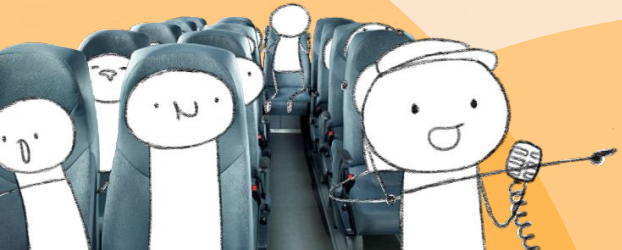


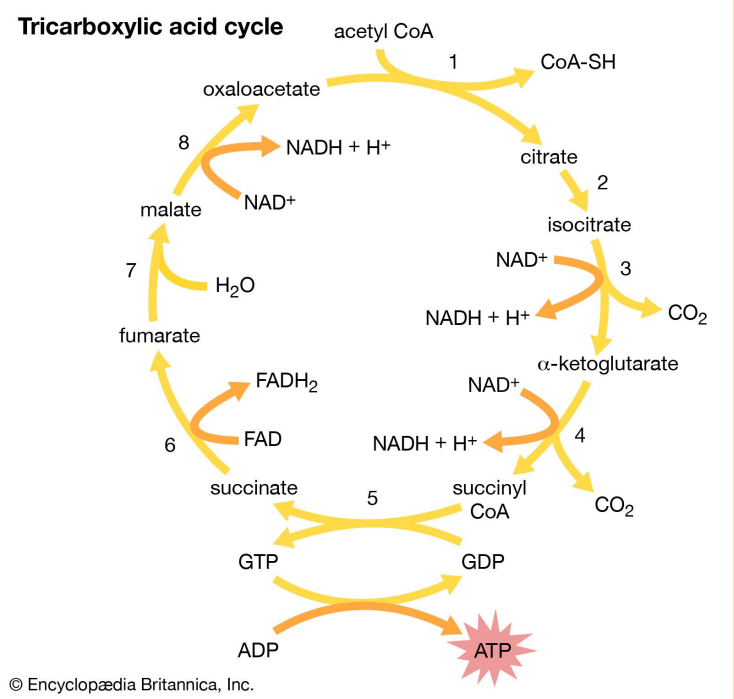
These are **redox reactions**, in which molecules are **reduced and oxidized**. For a molecule to get reduced, it gains electrons, while the oxidized molecule loses electrons. You can think of it like gaining electrons makes the formal charge more negative, thus reducing the charge. It's not a reduction of electrons, it's a **reduction of charges**! Of course, you can also just use the acronym **LEO says GER**, like a lion. LEO: lose electrons oxidized, and GER: gain electrons reduced. I'm a Leo, by the way. No I'm not, I'm a Virgo. Leos are so cool though, I wish I were a Leo.

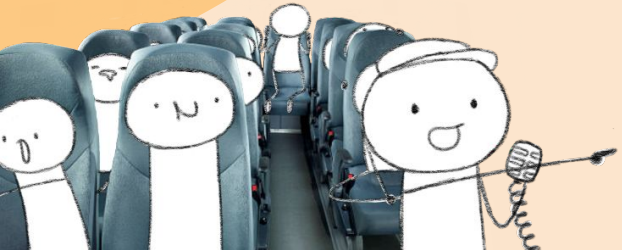




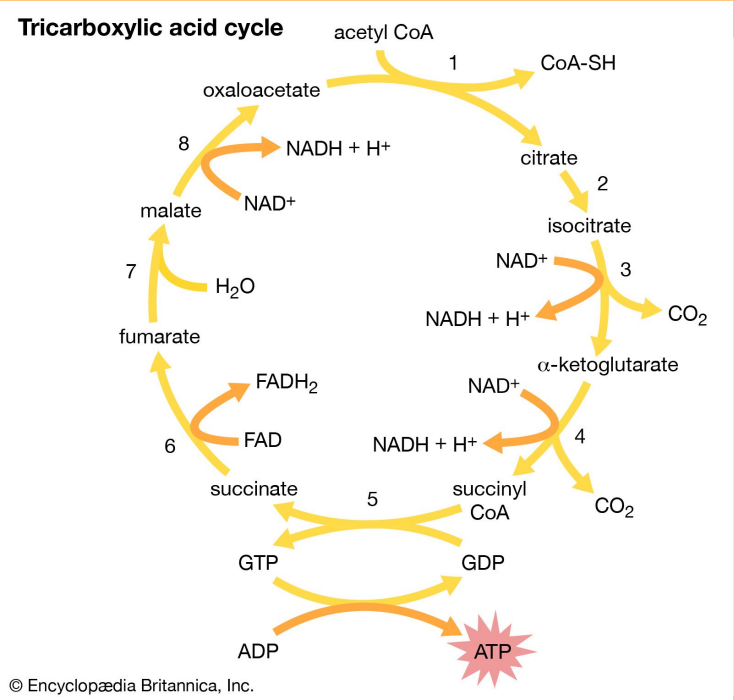
Pay attention everyone! We are currently entering the mitochondrion through a **transport protein**. Pyruvate oxidation is about to begin!

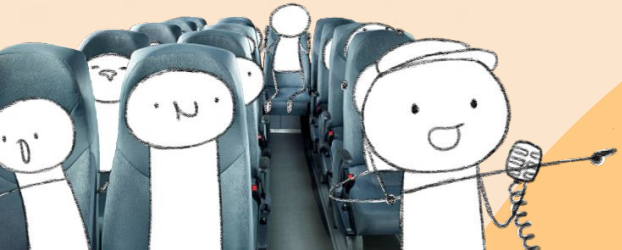
Now that we are in the matrix, the three-carbon pyruvate molecule will be converted into a two-carbon molecule attached to **Coenzyme A**, which we call **acetyl-CoA**. You can see an NAD+ molecule being reduced to NADH. We're dropping a carbon molecule as well, which is leaving as CO2. Here comes the Coenzyme A to attach onto us and... boom! This molecule is now acetyl-CoA! This bus is not a part of the chemical formula.





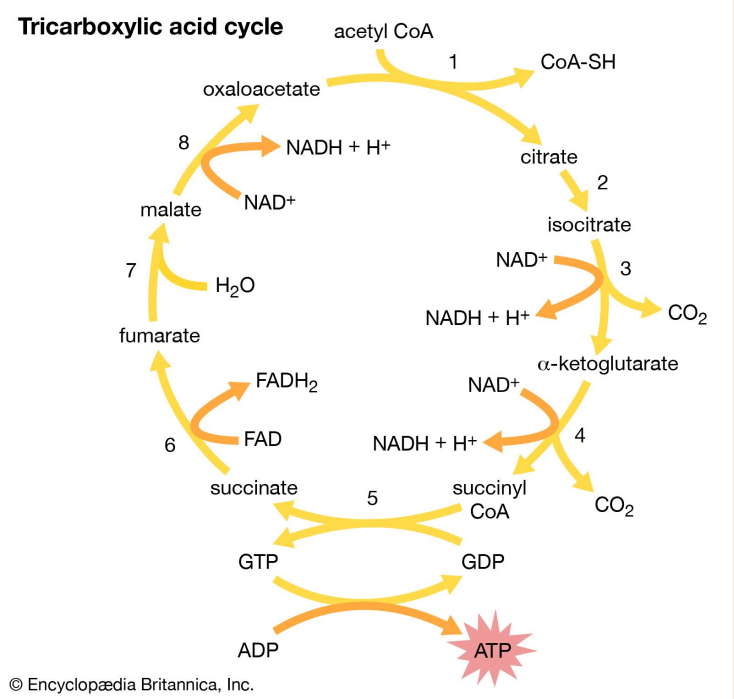
Now, this is where the Krebs Cycle is about to actually begin, as we enter the first step. Our friendly **Coenzyme A** is gonna transfer us and this acetyl group to begin the Krebs Cycle where we are gonna undergo a bunch of redox reactions! Things are getting reduced and oxidized, and losing and gaining electrons! For these redox reactions specifically, the electrons are transferred to the **electron carriers**, which are **NADH and FADH2**.

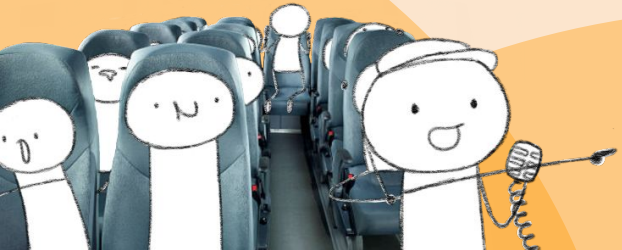




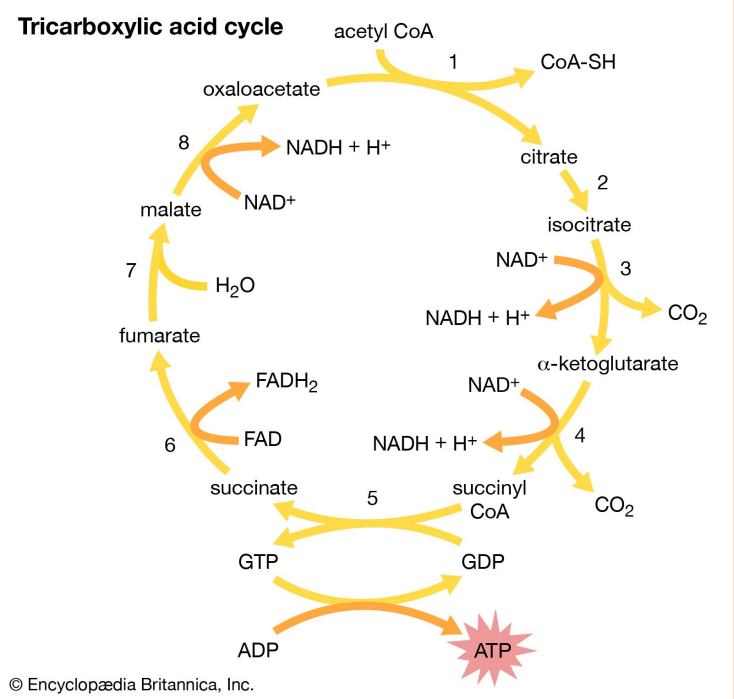
Right now, in the first step, we are being combined with **oxaloacetate** to produce **citrate**, with the help of our good enzyme friend **citrate synthase**. Right now, we've got 6 carbons!

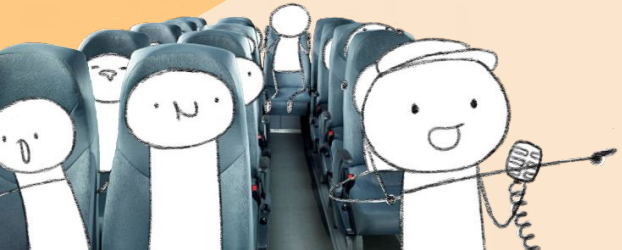
In step 2 we are converted into **isocitrate**, and this reaction is catalyzed by another enzyme called **aconitase**, and you can really begin to see how important of a role enzymes play in cellular respiration.



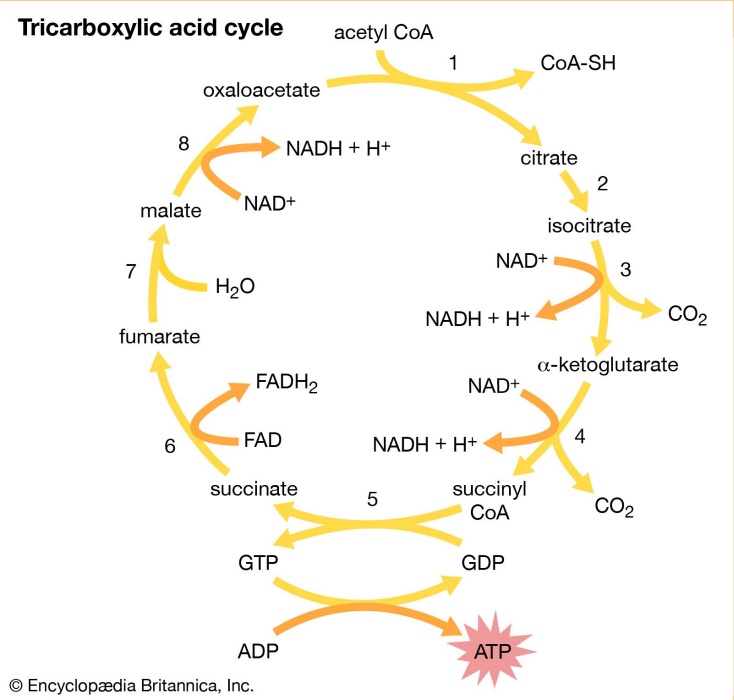


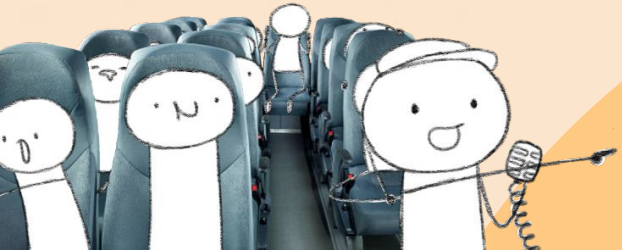
Next, we're gonna spice things up as the **isocitrate dehydrogenase** catalyzes this next reaction and we lose a carbon atom to CO2. Everyone say bye bye CO2! Look at them go! Here, NAD+ is also being converted to NADH. Don't blink or you'll miss it all!



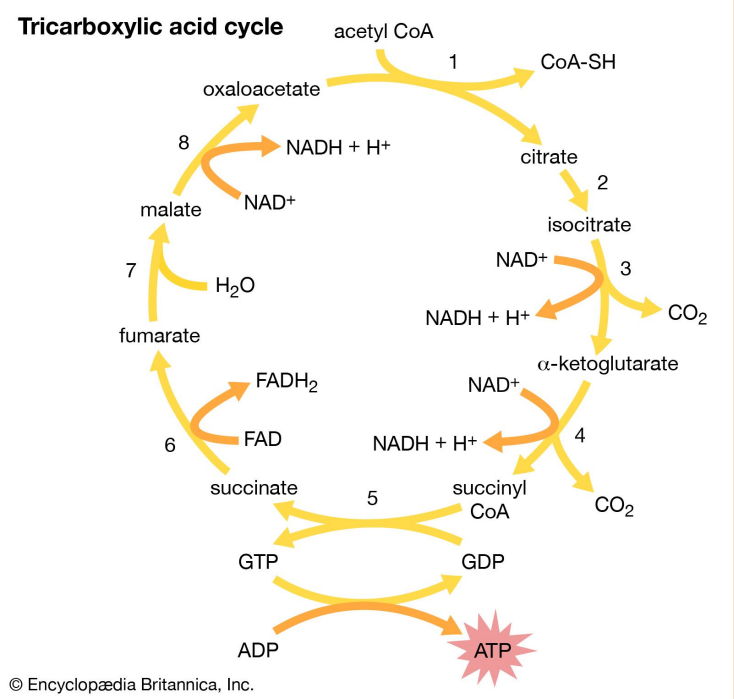


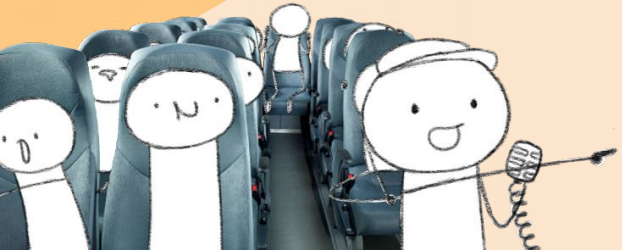
Moving on to step 4, we are now the 5-carbon molecule known as **α-ketoglutarate**. Did you think we were done? Nope! Here comes the **α-ketoglutarate dehydrogenase** enzyme, ready to catalyze the next reaction, in which we will be losing another carbon atom as they leave in the form of CO2. They grow up so fast. Another NAD+ is being converted to NADH, so it's very similar to what we had going on before!



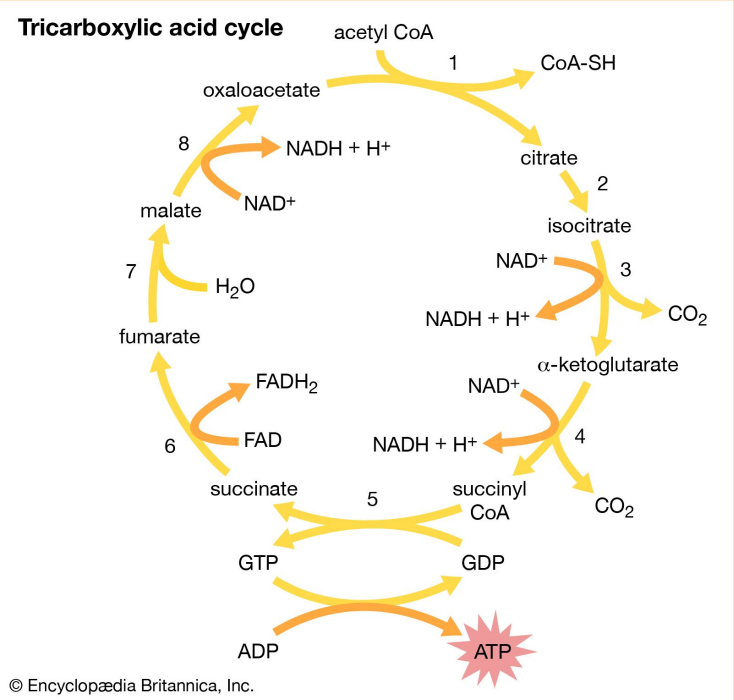


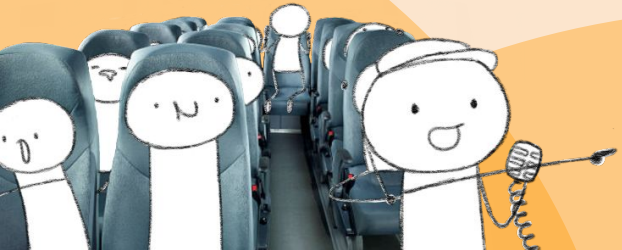
This molecule is called **succinyl CoA**, because of the Coenzyme A that is attached to us. The coenzyme is **replaced** by a phosphate group, but it's not going to stay with us forever! We are generously passing that phosphate onto the **GDP** and turning that "di" into a "tri", and now it's **GTP**! ATP and GDP are really the same thing, except that ATP has an **adenine** for A and GTP has a **guanine** for G. The entire reaction is catalyzed by the **succinyl CoA synthase** enzyme.





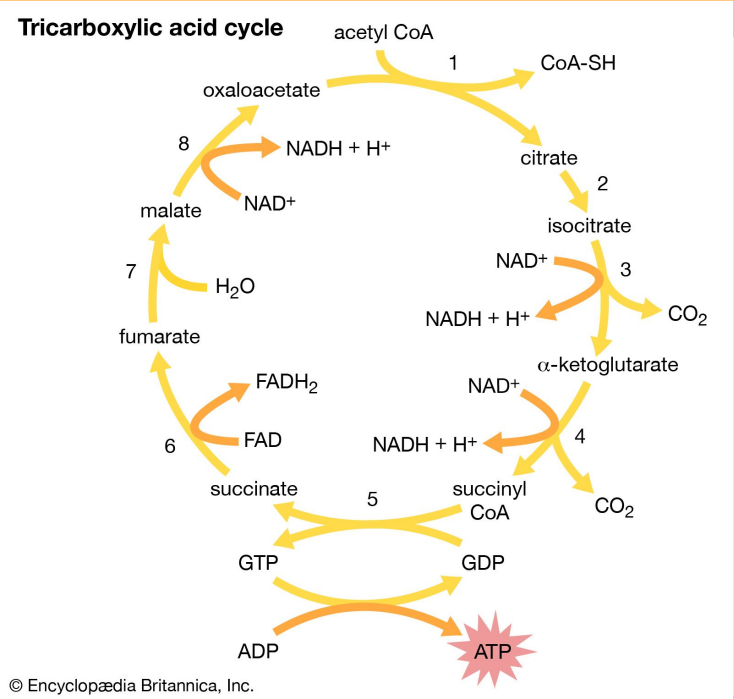
We're not done yet, though! Pay attention to that GTP molecule and that ADP molecule! The **nucleo-diphosphate-kinase** enzyme is helping transfer a phosphate group from the GTP to the ADP, to create **GDP and ATP**! In a way, ATP has been indirectly created in this step of the Krebs Cycle!

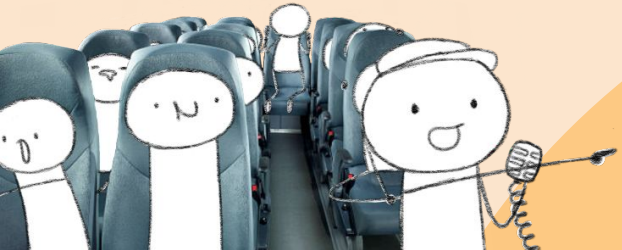




Now we enter step 6 as the 4 carbon succinate molecule! The reaction is going to be catalyzed by the **succinate dehydrogenase** enzyme, and FAD will be converted to FADH2, leaving us as the iconic **fumerate** molecule. All the molecules are iconic. So are you. You are also super iconic and cool.

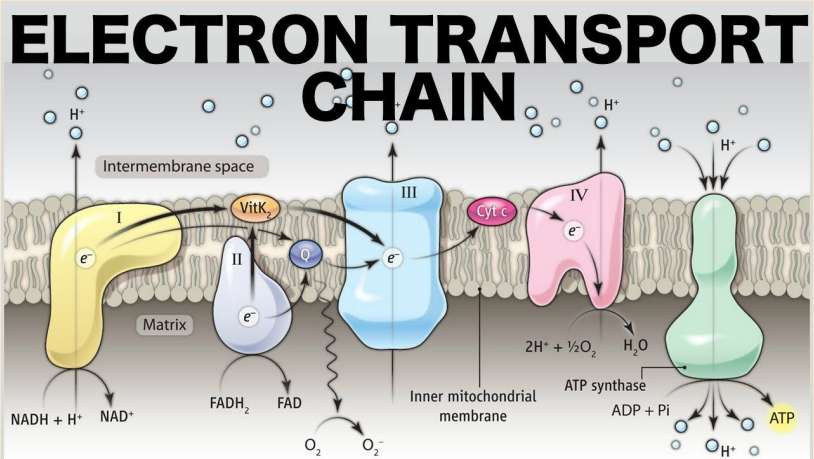
In step 7, the **fumerase** enzyme catalyzes our next reaction, in which a lovely, sweet water molecule joins us to form the **malate** molecule.

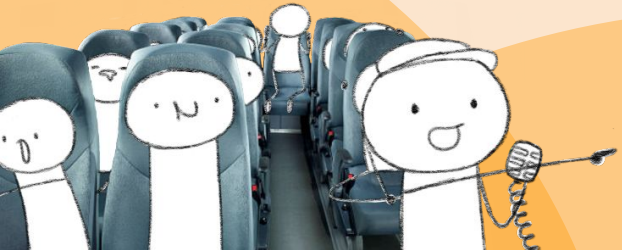




In our final step, step numero 8, the **malate dehydrogenase** helps the NAD+ steal a H+ ion from us to become NADH, and we've now been transformed into **oxaloacetate**, ready to help the next pyruvate molecule coming in to go through the Krebs Cycle, and we've successfully completed one full turn of the Krebs Cycle! Yeah!

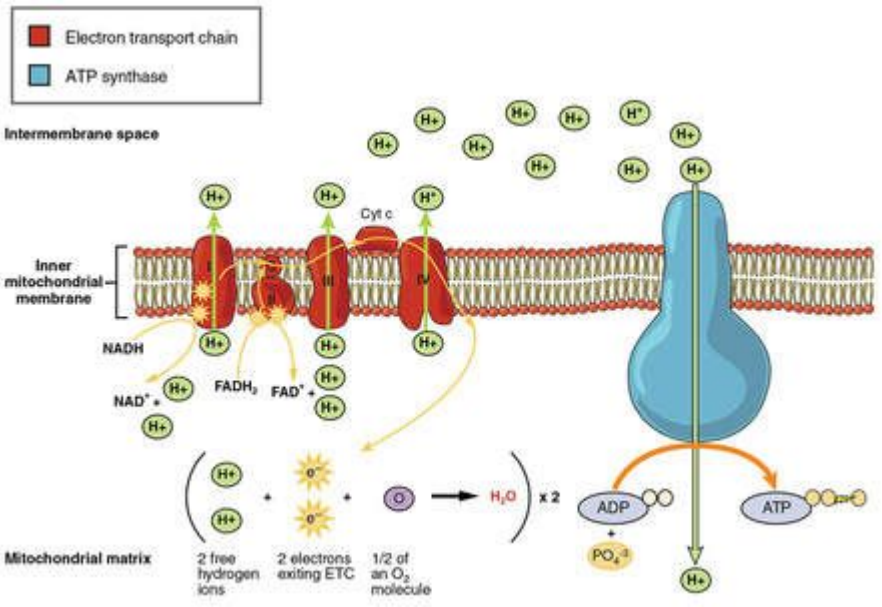
Everyone clap like the plane just landed, because that sure was one long journey!

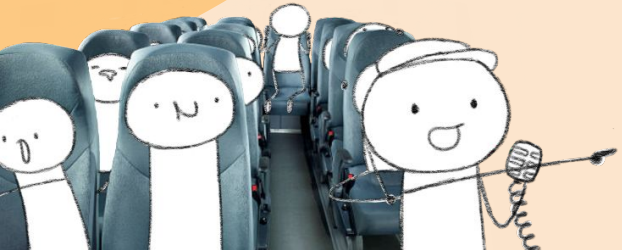




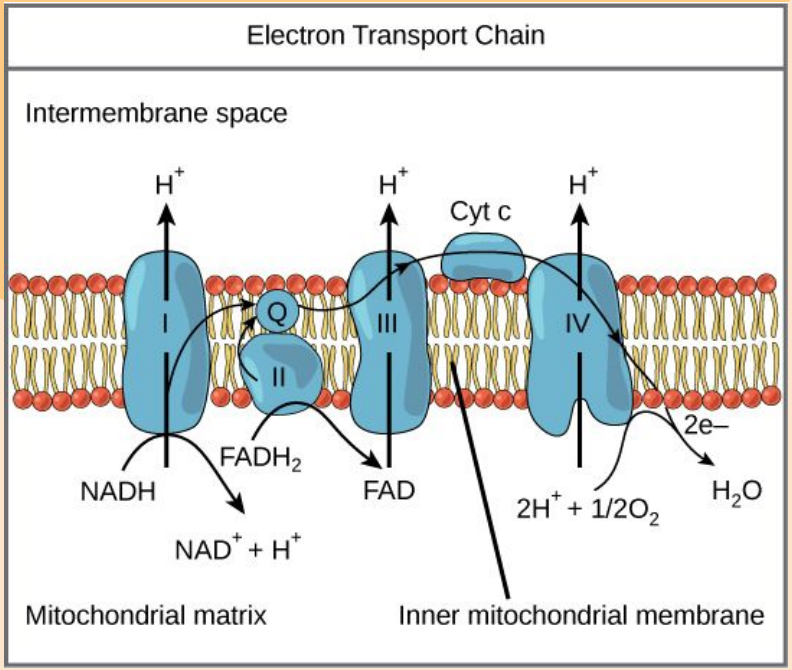
Now we're gonna sit through the Krebs Cycle again and slowly move onto an electron. This way, we're gonna get stolen away by NAD+ and join onto the newly produced NADH molecule to bring us to the **inner mitochondrial membrane**, where we'll watch the powerhouse that is the **electron transport chain** and the magic of **oxidative phosphorylation**!

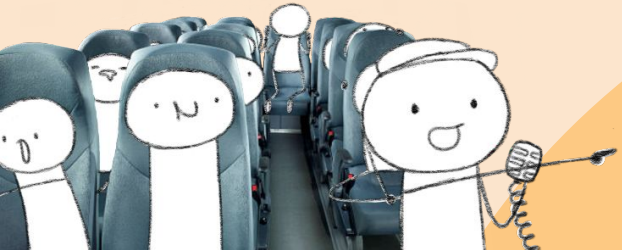
What is oxidative phosphorylation? It's an **aerobic process** and needs oxygen to function! This is because the energy from the electrons are going to be transferred to the final electron acceptor: **oxygen**!



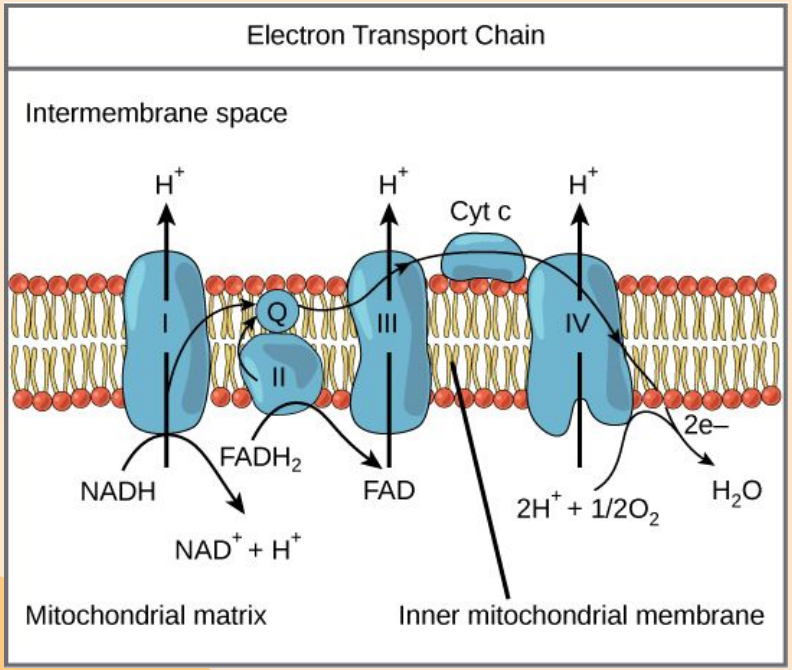


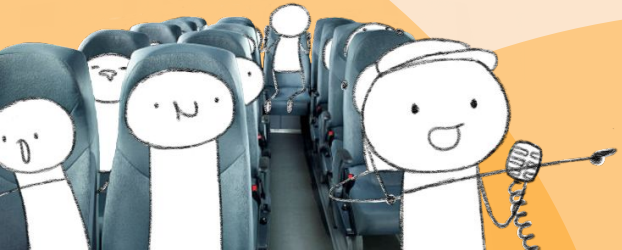
The electron transport chain is embedded in the **cristae**. Right now, the electron carrier molecules **deliver the electrons** to oxygen through a series of redox reactions. We are on the side of the membrane with a low concentration of H+ ions, while the other side, which is the **inner membrane space**, has a high concentration of H+ ions, making a **proton gradient**. The NADH is going to drop the electrons off here and leave for the matrix as NAD+.



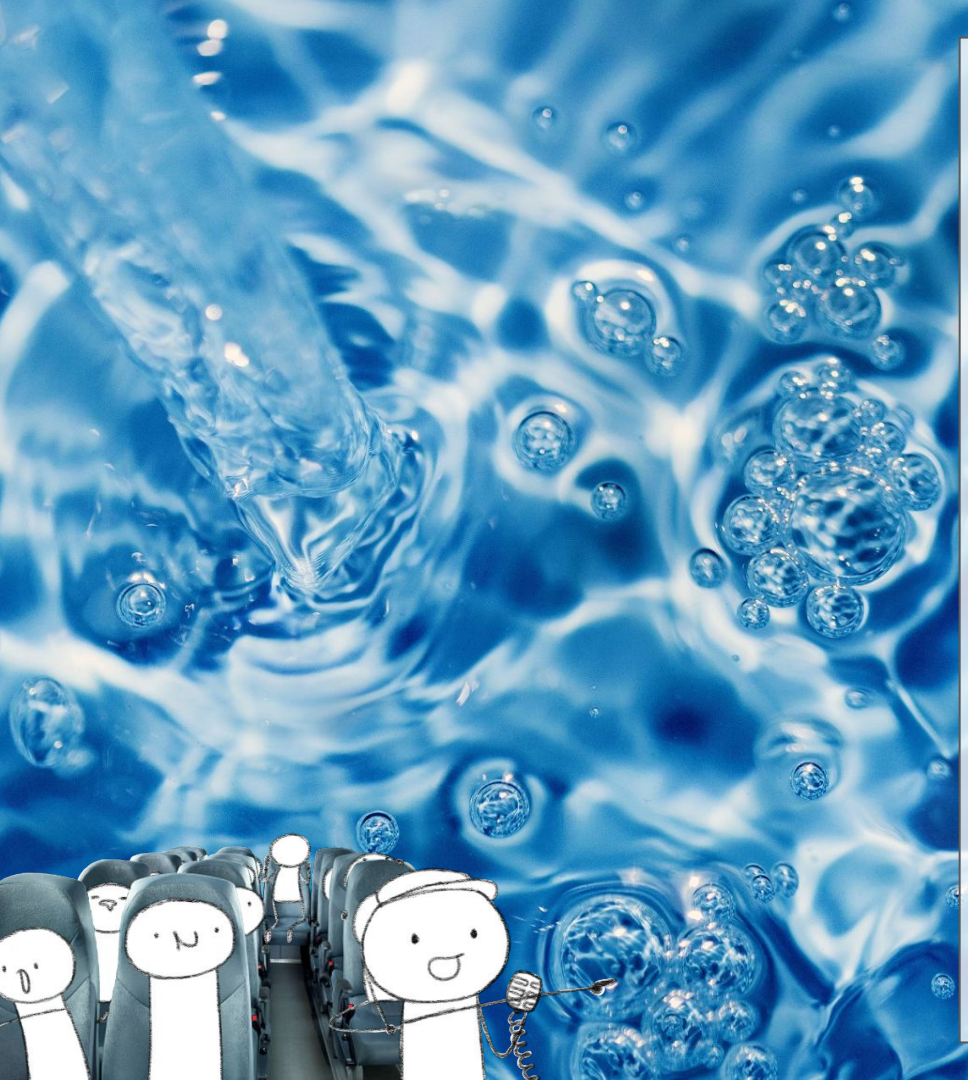


These proteins are called **carrier proteins**. That one is called **cytochrome-c**, and that one over there is the **coenzyme Q**, which can actually diffuse and move along the **fluid membrane**, which is what allows the chain to operate. Now, notice as the electrons are bouncing from protein to protein, they **release energy**, which **pumps** the protons, or the H+ ions, **across the membrane** to the inner membrane space, hence the name **protein pump**. They are being pumped **against** this concentration gradient as the other side has a higher concentration of protons.

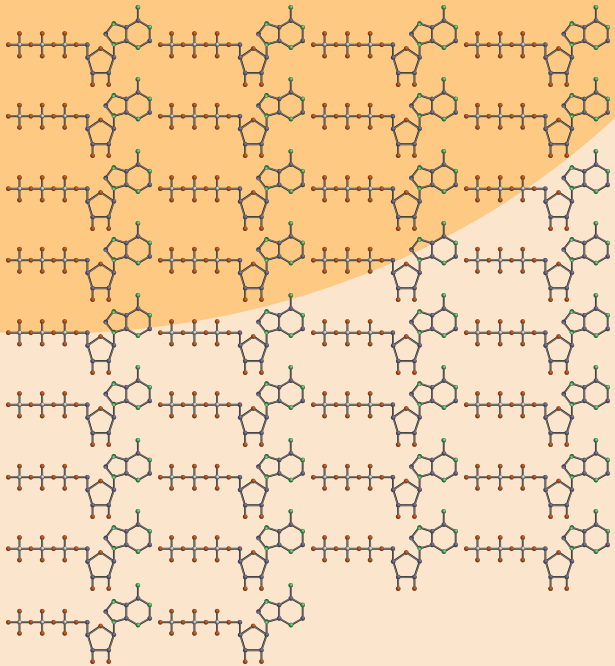




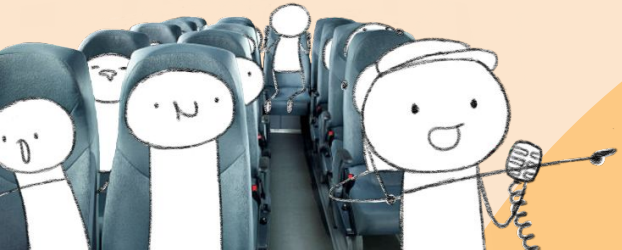
Over there, you'll see an enzyme called an **ATP synthase** embedded right there in the mitochondrial membrane. The H+ ions flow through that ATP synthase back to the matrix, which is the area of lower concentration. However, as the ions flow back in, they spin a **rotor**, and every time it rotates a phosphate is attached to ADP to form **ATP**, hence the phosphorylation part of **oxidative phosphorylation**.

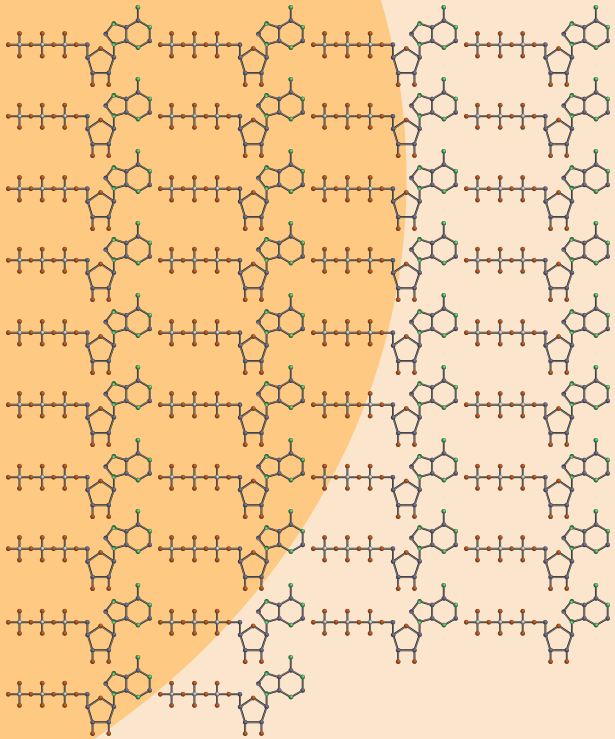


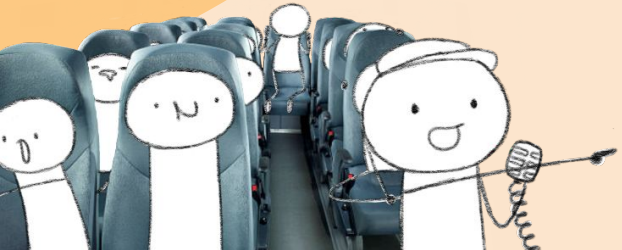
At the very end, the electrons and hydrogens will be combined with the oxygen to form **water**, which is why we unfortunately cannot actually hop aboard the electron transport chain. As fun as it may seem, there's no telling where that water molecule will end up and it will be quite a hassle to get back home.



Now remember how during glycolysis **2 NADH** were produced from 1 glucose molecule? And then that same glucose molecule produced 8 more NADH and 2 FADH2 during the Krebs Cycle? That brings us to a total electron carrier count of **10 NADH and 2 FADH2** that are travelling to the electron transport chain, all from 1 glucose molecule! **1 NADH produces 3 ATP**, and **1 FADH2 can produce 2 ATP**, so if you start working those cogs in your brain, you'll realize that it comes to a total of **34 ATP** being produced by **1 glucose molecule**.

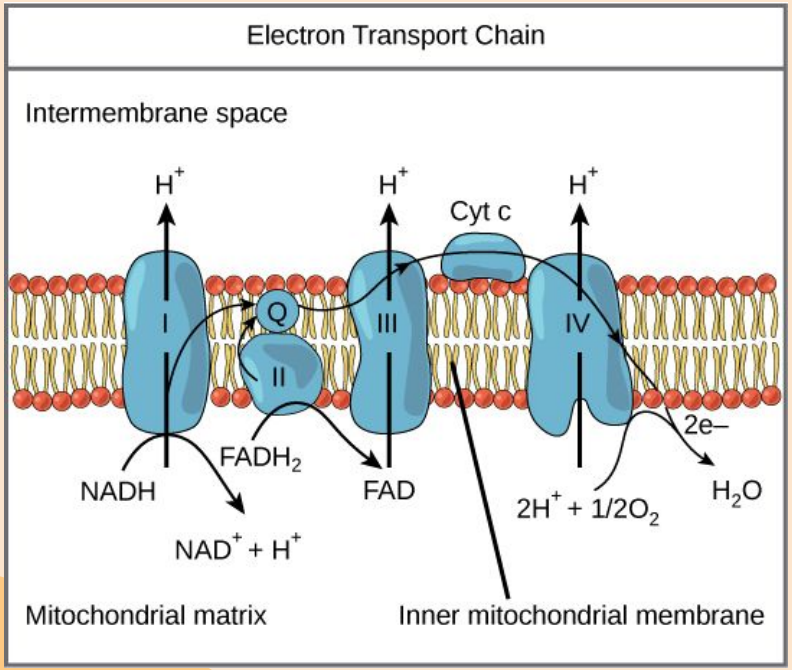


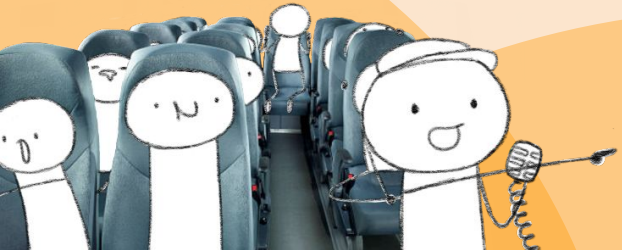




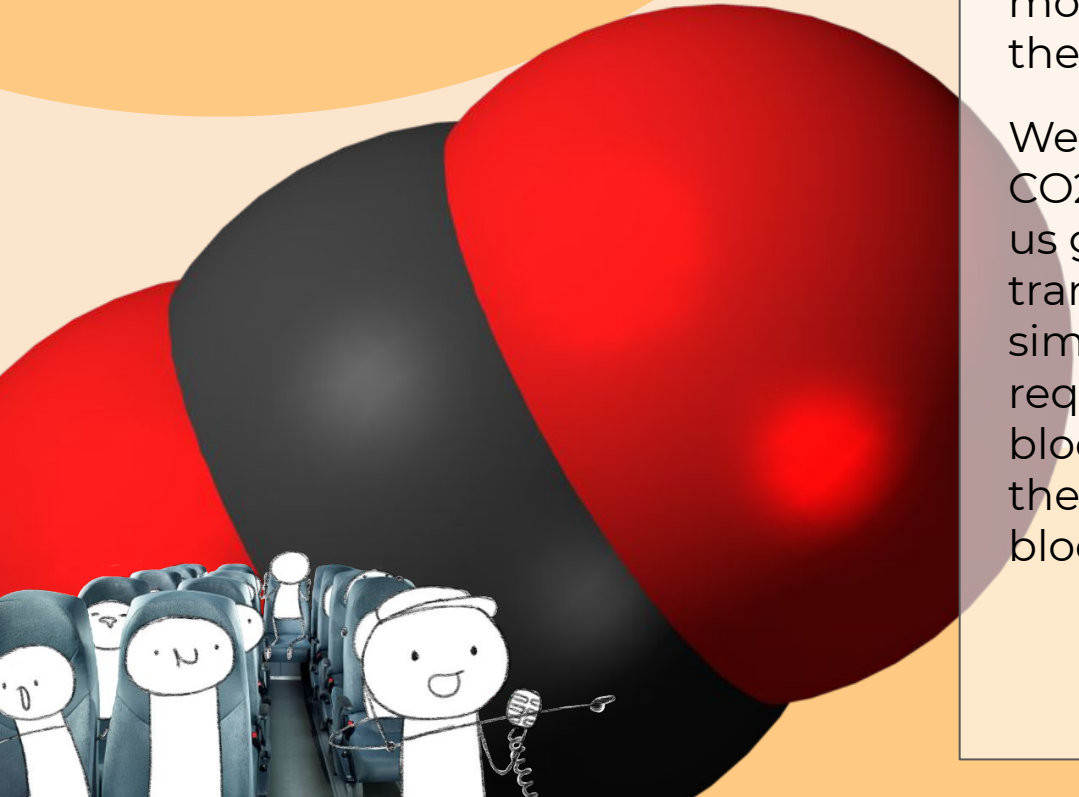
Afterwards, the ATP is transported out of the mitochondria and used to help **facilitate reactions** all over the cell. It's like the cell's fuel, and it'll be used for mechanical stuff, transport stuff, chemical reaction stuff, anything you can think of! This **ATP production** is the entire point of cellular respiration!

Remember that net gain of 2 ATP from glycolysis and the 2 more ATP produced in the Krebs Cycle? Not a lot compared to the whopping 34 ATP produced earlier! If we do all the math, we find that **one glucose molecule** produces **38 ATP** through the **total cellular respiration process**!



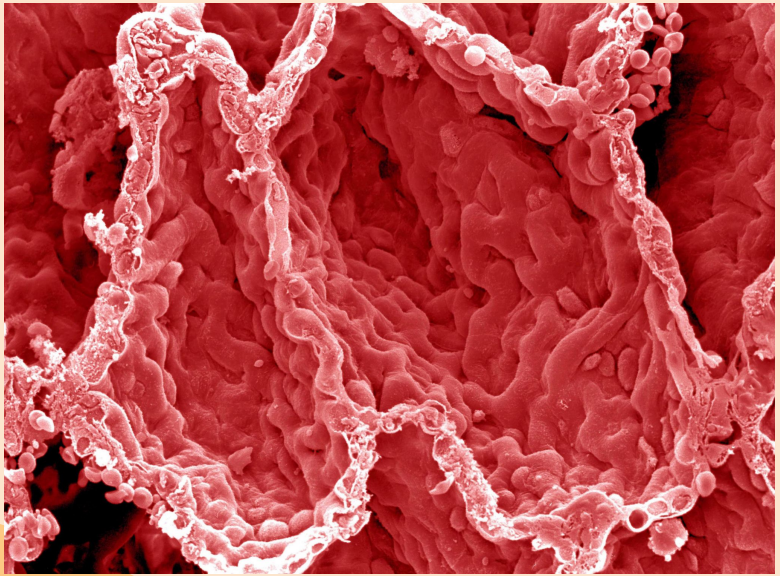


Coming back to oxidative phosphorylation, **oxygen** is what drives this entire process. Without oxygen, there would be no electron movement, and no proton gradient, and thus no ATP would be produced. Without movement in the electron transport chain, NADH and FADH2 cannot be oxidized to form NAD+ and FAD, which are the molecules we met back in the matrix during the Krebs Cycle, so it would also affect the functioning of the Krebs Cycle.

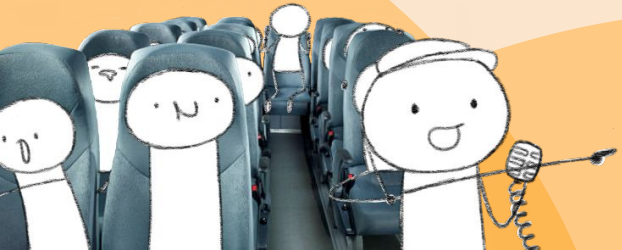


Now, us and the NAD+ are gonna head back over to the Krebs Cycle where we're gonna go find a CO2 molecule to bind onto so we can get the heck outta here!

We've found ourselves a helpful little CO2 molecule here who's gonna help us get back home. We're going to be transported across the membranes via simple **diffusion**, which doesn't require energy. Now we enter the bloodstream, where we're heading for the lungs. Buckle up y'all, the bloodstream does not play.



Well that was fun, wasn't it? The tour's almost over - all we've got left to do is diffuse into the **alveoli** in the lungs, get exhaled, and then pop right out into the outside world!



That’s all for today folks, I hope you enjoyed your tour and learned something new about your body and all the millions of tiny little guys living inside you working tirelessly, day and night, just to keep you alive. They’re working so hard breaking down your food and transferring electrons and phosphate groups, so be kind to them and help them out once in a while by eating healthy, staying active, and providing them with plenty of clean oxygen and strong nutrients so they can continue to do their jobs!